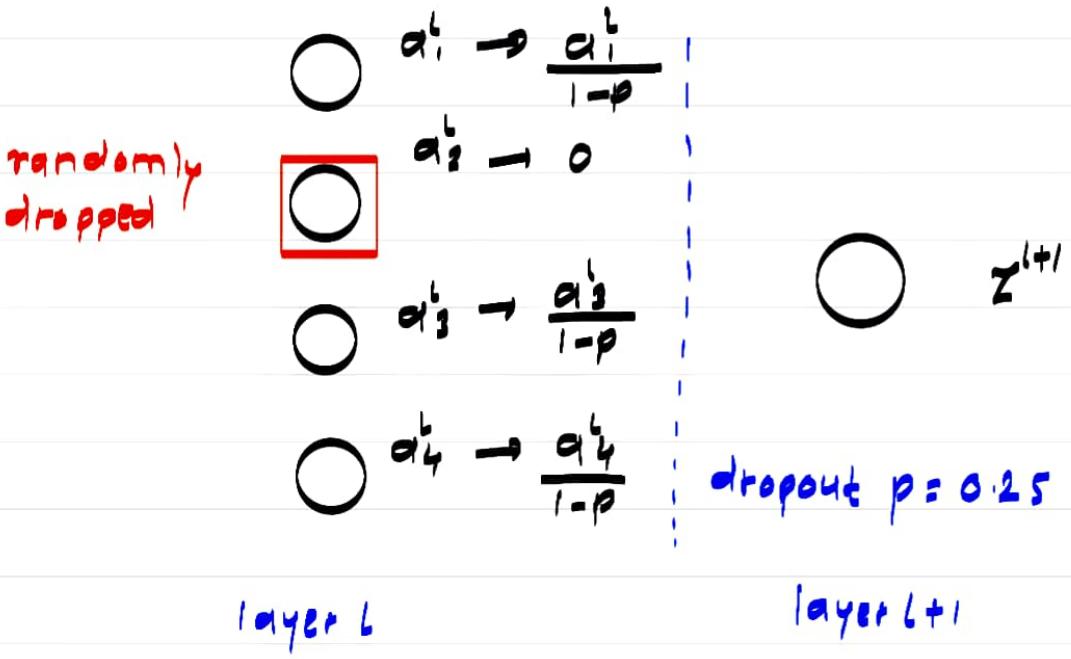
Deep learning

09-06-26

Dropout:

- regularization technique used in ANN to avoid overfiting
- its active only during training



$$z^{l+1} = w_1 a_1^l + w_2 a_2^l + w_3 a_3^l + w_4 a_4^l$$

z^{l+1} ↓ original value of z changes

a_2^l → 0

To reduce overfiting, we are not reducing no. of nodes but reducing no. of nodes which are involved in computation

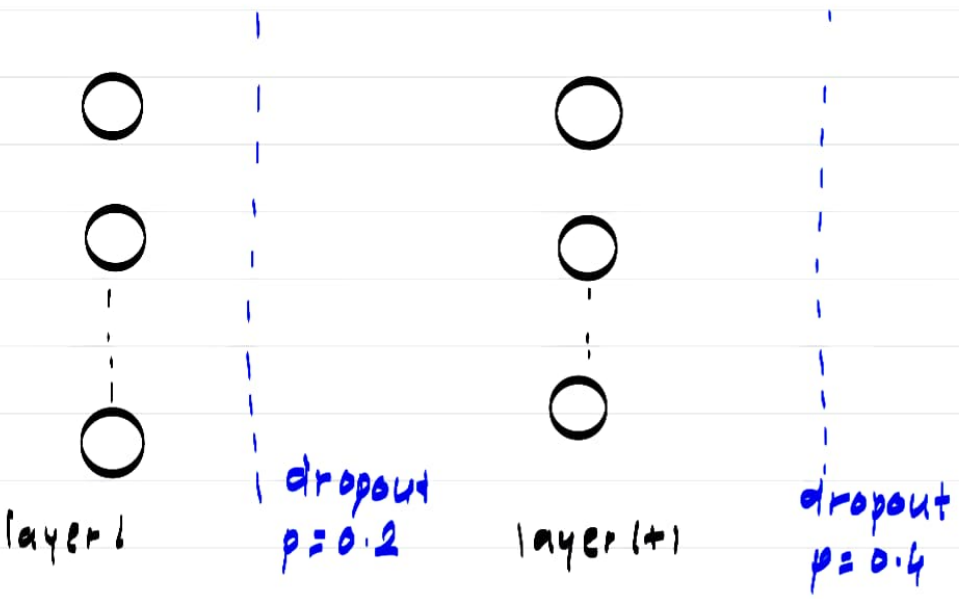
To adjust the scale of output, activations are multiplied by $\frac{1}{1-p}$ where p is dropout probability.

only multiplied during training

if n nodes are there in an ANN layer and $p = 0.25$ this means 25% of nodes in the layer will be dropped in dropout

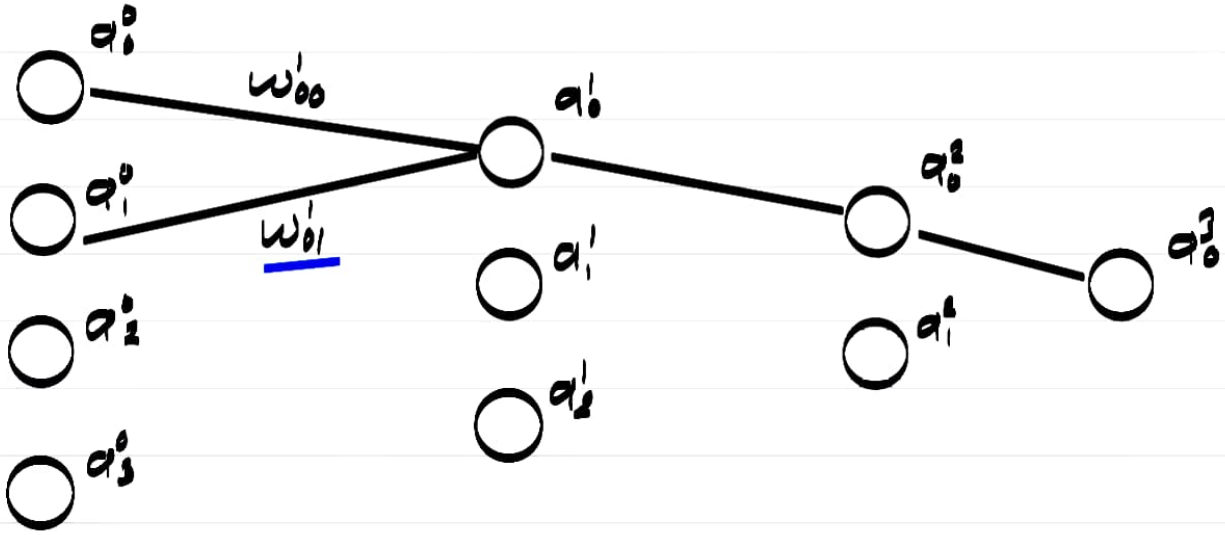
$\frac{n}{4}$	dropped no. of nodes
	1
8	2
12	3
20	5

Dropout is applied on a particular layer



During each forward pass p% of nodes are randomly dropped.

Normalization techniques in ANN:



input layer | layer1 | layer2 | output layer

feature scaling → normalized features

$$\frac{\partial L}{\partial w_{01}^1} = \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot \frac{\partial z_0^2}{\partial a_0^1} \cdot \frac{\partial a_0^1}{\partial z_0^1} \cdot \frac{\partial z_0^1}{\partial w_{01}^1}$$

$$= \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot \frac{\partial z_0^2}{\partial a_0^1} \cdot \frac{\partial a_0^1}{\partial z_0^1} \; a_1^0$$

$$z_0^1 = w_{00}^1 a_0^0 + w_{01}^1 a_1^0 + w_{02}^1 a_2^0 + w_{03}^1 a_3^0$$

$$\frac{\partial z_0^1}{\partial w_{01}^1} = a_1^0 \longrightarrow \text{one of the input features}$$

$$\frac{\partial L}{\partial w_{02}^1} = \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot \frac{\partial z_0^2}{\partial a_0^1} \cdot \frac{\partial a_0^1}{\partial z_0^1} \cdot \frac{\partial z_0^1}{\partial w_{02}^1}$$

$$= \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot \frac{\partial z_0^2}{\partial a_0^1} \cdot \frac{\partial a_0^1}{\partial z_0^1} \cdot a_2^0$$

- input features are directly involved in gradient computation
- if we don't scale them properly, features with higher scale will dominate training process
- hence we do feature scaling / normalization of features

$$\frac{\partial L}{\partial w_{01}^2} = \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot \frac{\partial z_0^2}{\partial w_{01}^2}$$

$$z_0^2 = w_{00}^2 a_0^1 + w_{01}^2 a_1^1 + w_{02}^2 a_2^1$$

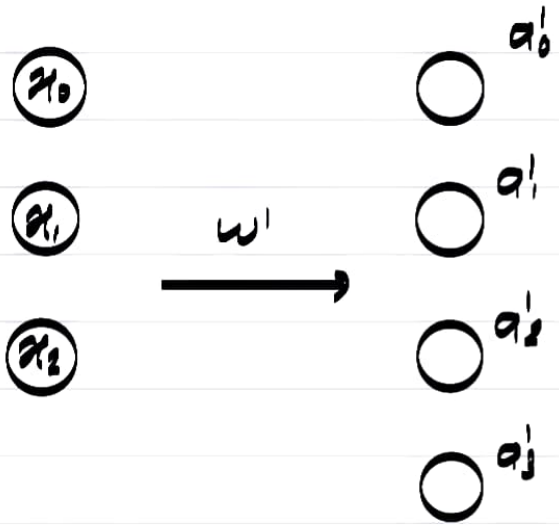
$$\frac{\partial z_0^2}{\partial w_{01}^2} = a_1^1$$

$$\frac{\partial L}{\partial w_{01}^2} = \frac{\partial L}{\partial a_0^3} \cdot \frac{\partial a_0^3}{\partial z_0^3} \cdot \frac{\partial z_0^3}{\partial a_0^2} \cdot \frac{\partial a_0^2}{\partial z_0^2} \cdot a_1^1$$

- As outputs of layers also being directly used in gradient computation, we need to normalize them

Layer Normalization:

- Used for normalizing layer level outputs
- in layer norm, normalization happens along the layer dimension.



layer 0 layer 1

data matrix $X = \begin{bmatrix} x_{01} & x_{02} & \cdots & x_{0n} \\ x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \end{bmatrix}$ $3 \times n$

$W^1 = 4 \times 3$

$Z^1 = W^1 X$

$4 \times n$

$a^1 = f_1(Z^1)$

$4 \times n$ $4 \times n$

$$a^1 = \begin{bmatrix} a^1_{01} & a^1_{02} & \cdots & a^1_{0n} \\ a^1_{11} & a^1_{12} & & a^1_{1n} \\ a^1_{21} & a^1_{22} & & a^1_{2n} \\ a^1_{31} & a^1_{32} & & a^1_{3n} \end{bmatrix}_{4 \times n}$$

$a^1_{01}, a^1_{02}, \ldots, a^1_{0n}$ are $d_1, d_2, \ldots, d_n$

we'll layer normalize these activations before passing to layer 2.

$$a^1 = \begin{bmatrix} a^1_{01} & a^1_{02} & \cdots & a^1_{0n} \\ a^1_{11} & a^1_{12} & & a^1_{1n} \\ a^1_{21} & a^1_{22} & & a^1_{2n} \\ a^1_{31} & a^1_{32} & & a^1_{3n} \end{bmatrix}_{4 \times n}$$

μ_1, σ_1 $\quad \mu_2 \sigma_2$ $\quad \mu_n \sigma_n$

$$\mu_1 = \frac{a^1_{01} + a^1_{11} + a^1_{21} + a^1_{31}}{4}$$

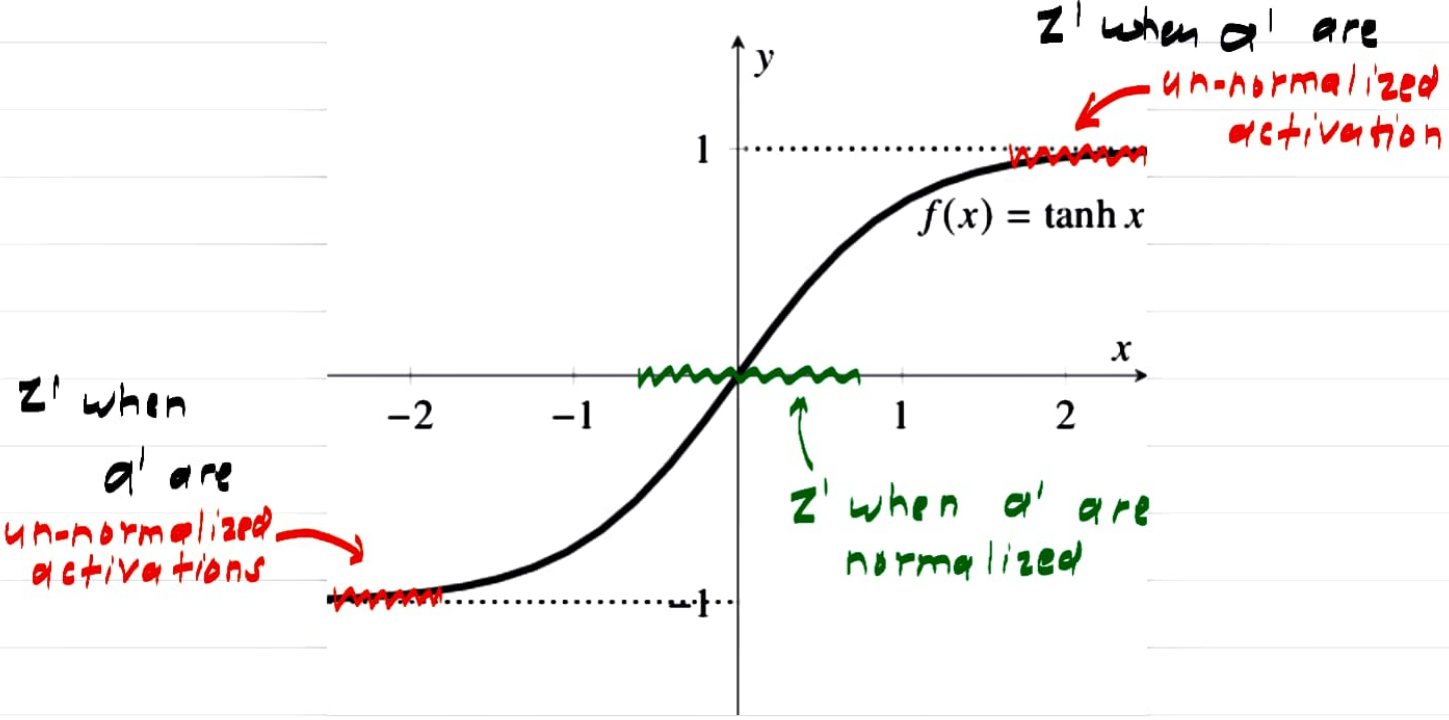
$$\sigma_1 = \frac{\sum_{i=1}^{4} (a^1_{i1} - \mu_1)^2}{4}$$

normalize activations as

$$a^1_{i1} = \frac{a^1_{i1} - \mu_1}{\sigma_1}$$

for any activation

$$a^1_{in} = \frac{a^1_{in} - \mu_n}{\sigma_n}$$



tanh is activation function in layer 2

$z^2 = w^2 a^1$

$a^2 = \tanh(z^2)$

as a^1 are un-normalized

$z^2 = w^2 a^1$ will be also un-normalized

Normalizing a^1 makes z^2 zero centered and zero centered z^2 will utilize full power of tanh activation function.

- this also avoids vanishing gradient issue.